

# An influence diagram approach to automating lead time estimation in Agile Kanban project management

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## ABSTRACT

This research introduces an influence diagram based approach to estimating task lead times for Agile Kanban project management. Derived from the principles of lean manufacturing, Agile methodologies including Scrum, Scrumban, and Kanban are common in the software industry and are spreading to other fields. Many teams estimate task delivery to better manage stakeholder expectations and improve decision making. However, the current technique involves calculating a team's story point completion velocity, requiring hours of weekly effort while overlooking the addition, removal, and reprioritization of backlog tasks. While these factors may be critical in Scrum, teams practicing Kanban experience these conditions frequently due to the emphasis on continuous integration and reprioritization. Our alternative approach applies an influence diagram, or Bayesian belief network, to model the uncertainties affecting the lead time. To partially automate the estimation process, an influence diagram based expert system is developed and populated with data from a practicing Kanban team and used to generate a cumulative distribution function to facilitate the communication of probabilistic estimates. A sensitivity analysis is conducted to better understand how each factor influences lead times. This system can assist Kanban teams' stakeholder communication and reduce estimation workload through a more holistic model of lead times.

## 1. Introduction

The explainability of decision and prediction-making expert systems has been a topic of research to improve user confidence in automated systems. An understanding of the factors that influence a prediction or recommendation, and the ability to conduct a what-if analysis of multiple scenarios, help a user better understand the system. The depiction of uncertainty in the predictions made by a system further helps the users of the system develop acceptable levels of confidence and reliance in the system. Bayesian theory has been used in expert systems to integrate uncertainty into the system output. A graphical model known as an influence diagram uses this mathematic model to create an explainable model based on Bayes theory and conditional probabilities. This research presents the application of an influence diagram based expert system to depict uncertainty in lead time estimation for Agile

Kanban project management. The goal of this study is not to improve the accuracy of the average lead time estimates. Instead, this practical implementation demonstrates the value influence diagrams can bring to an expert system by improving the user and stakeholder understanding of the system and by depicting the associated uncertainties in the system.

A key role of a project manager is to develop and control the schedules for their projects, which often involves calculating project delivery estimates used effective communication with project stakeholders (Institute, 2017). Project managers in the software industry, or other fields prone to rapid change or high levels of uncertainty have largely adopted Agile project management methodologies, and while the majority of teams practice Scrum, the newer process Kanban has been rapidly growing (VersionOne, 2019). A hybrid of Scrum and Kanban called Scrumban has also been gaining traction recently

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(Nikitina, Kajko-Mattsson, & Stråle, 2012). Due to Scrum's prominence, many Agile lead-time estimation tools have been developed with a focus on this methodology. The common process of estimating story points through planning poker requires the entire team to go through cycles of revealing their estimate of the effort required to complete a task, then reconcile the differences between team members through discussion (Grenning, 2002). This typically happens during either a multiple hour planning meeting in Scrum or the daily standup in Kanban. When story point estimation is applied to Agile Kanban—which can be thought of as a light version of Scrum that skips meetings outside of the daily standup—the level of overhead required to support estimation becomes even more disproportionate relative to other team administrative duties. This time-consuming process leads Agile teams to implementing more simplified but less accurate estimation techniques, such as “t-shirt sizing” of tasks, or to ditch estimation efforts altogether in an attempt to focus on adding value instead of spending time estimating. However, there is evidence linking the quality of estimation to project success (Trendowicz & Jeffery, 2014).

Organizational support is critical for the successful implementation of Agile (Conforto, Salum, Amaral, da Silva, & de Almeida, 2014). In addition, it is critical to choose a project management approach that matches the flexibility and structure of the project tasks in order to ensure project success (Kosztayán, Jakab, Novák, & Hegedűs, 2020). However, Kanban by design does not provide clear milestone dates for delivery like those provided by the detailed upfront planning and Gantt charts used by traditional waterfall project management. Instead, Kanban pulls the task through a workflow pipeline as resources become available. This absence of milestones can lead to a perceived lack of predictability, which can increase reluctance within an organization to transition teams to Agile methodologies, including Kanban (Hoda, Sal-leh, & Grundy, 2018).

When planning poker for story point estimation is applied to Kanban projects, the incorrect assumption is made that the project backlog will not experience change. In reality, Kanban continuously integrates tasks, allowing them to be added or removed from the backlog, or reprioritized at any time (PMI & AgileAlliance, 2017). Due to these challenges, more advanced analytical models should be developed in order to improve the estimation capabilities of Kanban teams, reduce time spent on non-value-added activities, improve the information communicated to stakeholders, and aid in the decision-making process of task prioritization (Dam, Tran, Grundy, Ghose, & Kamei, 2019). In order to quantify the multiple factors influencing the lead-time of a task, we implement an expert system using a model known as an influence diagram, also known as a Bayesian network or Bayesian belief network. Influence diagrams can be used to visualize the interactions between the factors that affect the lead time while bringing together both quantitative data and expert judgment to deliver a probabilistic estimate of lead time (Pendharkar, Subramanian, & Rodger, 2005). Our influence diagram visually represents the factors that affect the lead-time for a project task to leave the backlog and become work in progress (WIP). We developed our influence diagram through literature review and input from a professional Agile Kanban project team. The software Netica (Norsys Software Corporation) was used to develop an expert system based on this influence diagram, outputting probabilistic lead-time estimates in the form of a cumulative distribution function. This expert system was populated with data collected from a professional Kanban project team at our industry partner Andersen Corporation, a leading North American manufacturer of window and door products, as a demonstration of how the resulting information can be implemented to reduce the time team members spend producing lead-time estimates and how it can be used as a communication tool for stakeholders. This research devises an expert system based on a probabilistic model of task movement within the Kanban project backlog with the goal of reducing input required by the team and accounting for the factors that influence task movement, which are ignored by current estimation techniques.

In Section 2, we present an overview of Kanban that was used to

develop the structure of the influence diagram and a general mathematic description of influence diagrams. Section 3 covers the creation of the graphical influence diagram model, collection of Kanban team data, and creation of the interactive tool in Netica. In Section 4, the output of the model in the form of a cumulative distribution function is presented and discussed, along with a sensitivity analysis of the five factors that influence the task lead-time. Section 4 also presents a comparison between current estimation techniques and the estimates produced using the influence diagram based expert system. Last, Section 5 summarizes the conclusions made during the development of this estimation tool and looks forward to the next steps into further automating Agile Kanban project management and other applications of influence diagrams in expert systems.

## 2. Background and related work

### 2.1. The Agile Kanban process

Managing complex projects like the development of a new product is difficult because the tasks have variable levels of uncertainty, and requirements may not be known or constant (Parnas & Clements, 1985). To help manage this, the project management methodology of iterative product development began emerging in the 1930s. These concepts have been applied heavily in software development and became popular through the publication of the Agile Manifesto, a set of guiding philosophies for how to manage the dynamic nature of projects in that industry (Beck, Beedle, Bennekum, Cockburn, Cunningham, Fowler, & Thomas, 2001; Larman & Basili, 2003).

Iterative project management and the Agile principles were used to guide the evolution of a variety of project management methodologies, including Scrum, Dynamic Systems Development Method (DSDM), Extreme Programming (XP), Kanban, and many more (Misra, Kumar, Kumar, Fantasy, & Akhter, 2012). Kanban evolved from Scrum at Microsoft and Corbis, and it is described in detail in Anderson (2011) book, *Kanban*, and Eric Brechner (2015) book, *Agile Project Management with Kanban* among others. Both Kanban and Scrum visualize a team's workflow and the project tasks planned, in process, and completed through the use of a physical or digital board. These tasks are sometimes referred to as user stories since they often represent a user's requirement or requested feature that the team will deliver (Brechner, 2015). Fig. 1 shows an example of a Kanban board that a team might use to visualize their tasks and workflow. Planned tasks are contained in the Backlog column, and active WIP is pulled through the workflow pipeline from left to right, which in this example consists of the three columns: *Break Down*, *Improve*, and *Document*. While we show these workflow steps here for demonstration purposes, each team identifies the steps to their unique workflow, changing the labels and number of columns to fit their process (Brechner, 2015). Finally, completed tasks are placed in the final column labeled *Done*.

Currently, the leading technique for estimating Scrum and Kanban project delivery is a Wideband Delphi method known as planning poker, which assigns point values to each task, or user story, in the backlog that

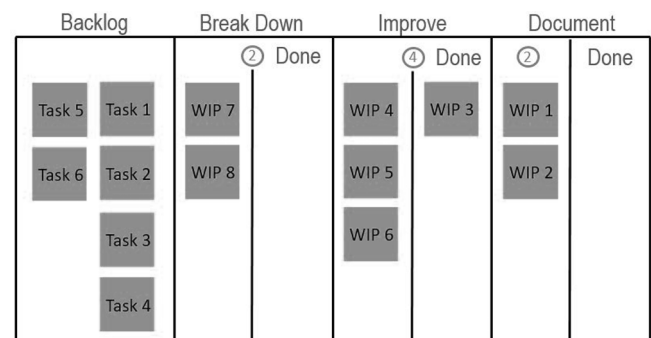


Fig. 1. Example Agile Kanban board populated with tasks and WIP.

represents the relative volume of work required for completion (Mahnić & Hovelja, 2012). These points are commonly referred to as story points (Brechtner, 2015). The number of story points completed in a certain time period by the team is referred to as the team's "velocity," with units of story points per day. Each team will have a best case, worst case, and average velocity which is used to create estimates for the minimum, maximum and average number of story points worth of work the team can be expected to complete in a given time period going forward (Downey & Sutherland, 2013). However, this estimation technique may be frustrating for Kanban teams to apply since estimates will be inaccurate as soon as the backlog changes, which can happen daily since this technique assumes no backlog changes occur. To address the inherent inaccuracy of these estimates, a more advanced analytical technique is needed that can account for the task's movement in the backlog.

The term Kanban has been used for many decades in lean manufacturing in reference to the just-in-time (JIT) production control system developed by Toyota that is driven by production line demand instead of using work in progress (WIP) buffers and production schedules (Sugimori, Kusunoki, Cho, & Uchikawa, 1977). Kanban is a "pull" style of material delivery because parts are withdrawn from a preceding step in the process as demanded by the assembly line. In contrast with "push" systems where parts are produced and delivered based on a forecasted demand schedule (Sugimori et al., 1977). While using Scrum, Anderson (2010) noticed that the workflow was "push" based, where tasks would move to the next step as soon as they completed the previous step leading to excessive WIP. The concept of Kanban with its JIT "pull" flow was used to evolve the Scrum Agile project management process into what is now known as Kanban with the goal of achieving more sustainable workflow and engraining a kaizen culture of continuous improvement (Anderson, 2010). This continuous improvement mentality and the openness of Agile practitioners to experiment is exemplified by the development of the hybrid methodology Scrumban, which uses the best features of both methods in order to avoid the weaknesses of each (Reddy, 2015).

In Kanban, the backlog is managed in a different way than in Scrum. Scrum uses finite intervals of time referred to as time boxes or sprints, at the beginning of which a prioritized sprint backlog is populated with project tasks pulled from a project backlog. The project tasks are planned in detail and held mostly constant for the duration of the sprint (Downey & Sutherland, 2013). Conversely, Kanban does not use sprints, with tasks instead being pulled in a continuous flow from the top of a single prioritized backlog list maintained by the team regularly during daily standup meetings. The lack of a locked-down sprint backlog in Kanban leads to a dynamic backlog that is frequently being re-organized. The dynamic backlog is considered by practitioners as an advantage of Kanban, providing more flexibility or agility, and since work has not been started on those tasks yet, the productivity of the team is not impacted (Brechtner, 2015). However, the estimates provided to stakeholders are out of date nearly immediately when following the current techniques developed for Scrum's more stable backlog, since they do not account for this variability.

In order to provide accurate lead time estimation in Kanban, we must account for the factors that influence how a task will flow through, and eventually exit the backlog. First, there is typically a steady stream of new tasks added to the backlog ordered by priority (Sedano, Ralph, & Péraire, 2019). There may also be cases where a project task that was submitted to the backlog will no longer be required by the customer, in which case, those items will be removed by the team. The order of project tasks in the backlog can also be changed by the project team at any time (Brechtner, 2015). Without taking these variations in the backlog into account, any estimate provided to the customer about when project tasks will be completed will be inaccurate.

Many project teams are expected to provide estimates to their customers, leadership, or partners for when tasks in the backlog will be delivered (Brechtner, 2015). Since each item in the backlog will take a different amount of time to complete, and there are often unknowns

related to the work content, estimates are challenging to produce (Anderson, 2010). In addition, the level of productivity within a project team is not constant due to variables such as sick and vacation leave, non-project meetings, and non-project administrative tasks. If an estimation technique is going to be accurate, it must also account for these variables.

## 2.2. Expert systems and Bayesian networks

For many decades, researchers have developed and implemented rule-based systems that utilize a knowledge base and inference system to replicate the judgment of a professional, in what is known as an expert system (Jensen, 1996). Normative expert systems support the user in their decision-making process by quantitatively modeling the domain through the application of probabilistic methods such as Bayesian Networks (Pearl, 1986). By identifying the variables or sources of information that have a direct or indirect influence on the outcome that is being modeled, a visual representation in the form of a directed, acyclic graph can be generated consisting of nodes that denote the variables and arrows representing dependency (Charniak & Goldman, 1993). These graphs are referred to by a number of names interchangeably including influence diagrams, Bayesian networks, or belief networks.

A Bayesian network or an influence diagram models decisions and uncertainties that contribute to one or more final outcomes (Howard & Matheson, 1984; Pearl, 1988). The influence diagram uses conditional probabilities to relate the uncertainties together and results in a probability distribution of the final outcome. Decisions are included in the influence diagram to help a decision maker understand how the probability of an outcome is influenced by each alternative (Shachter, 2007). For example, Fig. 2a depicts an influence diagram with a decision D (a rectangle), two uncertainties A and B (circles), and a deterministic outcome O (a rounded rectangle). As shown by the arrows, the uncertainties A and B depend on the decision D and the outcome O depends on the uncertainties A and B. Fig. 2b constructs the same influence diagram using Netica software which depicts the number of alternatives within the decision and the number of states in uncertainties and outcome. Decision D has two alternatives d\_1 and d\_2; uncertainty A has two states a\_1 and a\_2; uncertainty B has two states b\_1 and b\_2; and outcome O has two states o\_1 and o\_2. The probabilities depicted in the uncertainties and outcome indicate that alternative d\_1 is selected.

Computing the probability of o\_1 and o\_2 for outcome O requires several assessments. First, it is necessary to assess the probability of a\_1 and a\_2 for uncertainty A and the probability of b\_1 and b\_2 for

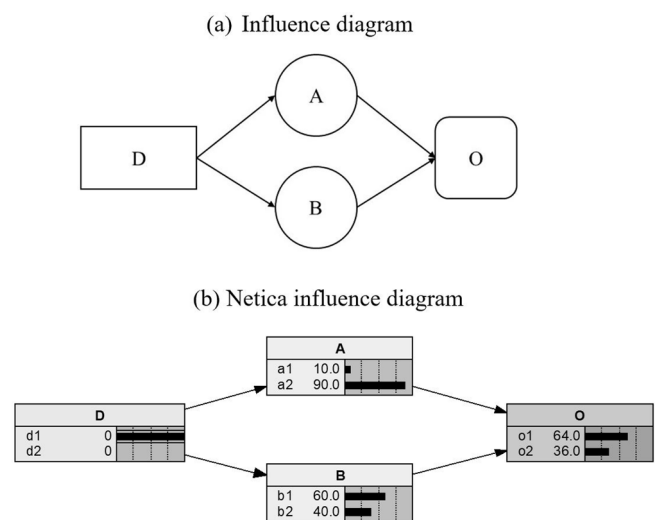


Fig. 2. An influence diagram with (a) the structure only and (b) the probabilities constructed in Netica if alternative d1 is selected. The influence diagram has one decision D, two uncertainties A and B, and one deterministic outcome O.

uncertainty B. Fig. 2 displays the probabilities of A conditioned on D = d<sub>1</sub>:  $P(A = a_1 | D = d_1) = 0.10$  and  $P(A = a_2 | D = d_1) = 0.90$ . The probabilities conditioned on D = d<sub>1</sub> are:  $P(B = b_1 | D = d_1) = 0.60$  and  $P(B = b_2 | D = d_1) = 0.40$ . Since the outcome O is deterministic conditioned on A and B, it is necessary to define which combinations of A and B determine o<sub>1</sub> and o<sub>2</sub>. In this example, O = o<sub>2</sub> if and only if A = a<sub>2</sub> and B = b<sub>2</sub> and O = o<sub>1</sub> for the other combinations. These probabilities and relationships are typically assessed through a combination of data and expert elicitation (Jensen, 1996). In Fig. 2, the probability that O = o<sub>2</sub> given that d<sub>1</sub> is selected equals the probability that A = a<sub>2</sub> and B = b<sub>2</sub>:

$$\begin{aligned} P(O = o_2 | D = d_1) &= P(A = a_2 | D = d_1) * P(B = b_2 | D = d_1) \\ &= 0.9 * 0.4 \\ &= 0.36 \end{aligned}$$

Since the probabilities of A and B conditioned on decision D are independent of each other, the probability of both is a product of each. Software such as Netica quickly calculates these joint and marginal probabilities after the assessed probabilities are entered into the model.

Influence diagrams have been popular modeling tools for analyzing the risks of engineered systems (Lindley, Bedford, & Cooke, 2002; Trucco, Cagno, Ruggeri, & Grande, 2008), decision making in business and public policy (Neil, Fenton, & Tailor, 2005; Pat-Cornell & Guikema, 2002; Varis, 1997), and diagnosing disease (Owens, Shachter, & Nease, 1997). Influence diagrams typically are constructed from collected data as well as subject matter expertise to assess uncertainties for which data is not available (Agogino & Rege, 1987; Neil, Fenton, Forey, & Harris, 2001; Neil et al., 2005; Stallard (Voelker), MacKenzie, & Peters, 2018). Recently there has been interest in applying Bayesian network expert systems to improve estimation in project management. Bayesian networks have been used to create expert systems to manage risk in large projects (Lee, Park, & Shin, 2009). Perkusich, Soares, Almeida, and Perkusich (2015) created a Bayesian network system for the evaluation of the robustness of Agile Scrum project management processes. A Bayesian network is used to evaluate project risks that affect the timeline of a construction project (Luu, Kim, Tuan, & Ogunlana, 2009). A recent study compared the effectiveness of a Bayesian network in predicting task effort in software development to that of a nonparametric neural network and a regression tree forecasting model. The accuracy of the Bayesian network is competitive to the other methods in providing single point estimates, but the Bayesian network has the additional benefit of providing probabilistic bounds (Pendharkar et al., 2005).

### 3. Influence diagram for lead time estimation

#### 3.1. Modeling approach

This study presents a model to assist Kanban teams to provide an accurate estimate of when work would start on a specific backlogged task by taking into account factors such as team performance, task sizes, and movement in the backlog. In this section, we discuss the development of an influence diagram that represents the relationships between these elements. Every Kanban project team has a unique workflow that a task will flow through once it leaves the backlog and becomes WIP, making it difficult to create a general purpose model for these steps (Brechtner, 2015). Once work on a task has started, it likely becomes easier for teams to provide estimates that are accurate because the work content of the task is better understood. Because of this, the specific workflow for this project team was intentionally left out of this model so that it can be applied to a broad range of project teams, lowering the effort required for teams to adopt this tool. Kanban aims to be a simple, easy to implement Agile methodology. In this spirit we present a straightforward, easily implemented tool created with the goal of it being applicable to a broad spectrum of project teams.

#### 3.2. Model development

The model is generated by first identifying the six factors that help determine the true nature of how a task will progress toward exiting the backlog: (i) the task's initial backlog position, (ii) new tasks added to the backlog, (iii) tasks canceled from the backlog, (iv) reprioritization of backlog tasks, (v) the effort required to complete the backlog tasks measured in story points, and (vi) the rate at which a team is completing work measured in story points per day, often referred to as velocity. These factors were identified in our review of the literature and were evaluated by a Kanban project team at our industry partner Andersen Corporation (Bayport, Minnesota US). Links are formed between these nodes to represent how they influence each other, leading to the acyclic graph shown in Fig. 3. Following the procedure outlined in Section 2.2, this graphical model will be transformed into a mathematic model consisting primarily of conditional probability calculations associated with the dependencies identified in Fig. 3. Further reading on the formal mathematical model of influence diagrams can be found in the referenced research presented in this section. In Section 4.2, the calculated conditional probabilities for each node are presented in table form. The output of the model will be a probabilistic estimate of the time required for the task to move from its starting position in the backlog to the backlog exit where it will become WIP. This lead time will be influenced by and inform the team's decision of where to initially position the task within the backlog.

##### 3.2.1. Backlog position

The first node in this diagram, the rectangle labeled *Backlog Position* on the left side of Fig. 3, is the position the team elects to place the task in the project backlog, which is the ordered list of all tasks in a project on which the team has not yet started work. The team decides where to position a task within the backlog based on the relative importance of the new task in comparison to the existing tasks. The relative importance is in part based on the value that the task's completion will deliver to the customer. Since Kanban allows the project team to reprioritize tasks in the backlog at any time, the team can decide to change the value of this node. A situation where a task may deliver customer value if delivered before a specified date or reduce value if delivered after that deadline is an example of a time where the team may be able to increase customer value by changing the task's position based on feedback from this model.

The *Backlog Position* node is connected to four uncertainty nodes (circles), as seen in Fig. 3, indicating the probabilities of the states of these nodes are conditional on the alternatives in the decision node. For example, if a task prioritized as the tenth most important task, there will be more opportunities for new tasks to be added at a higher priority than there would for a task ranked fifth in the backlog. Thus the probabilities of each state in the node *New Tasks Added to the backlog* will be different for these two scenarios.

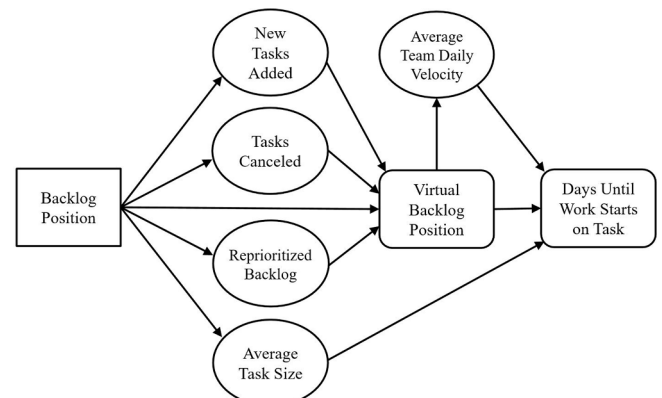


Fig. 3. Bayesian network representing factors that influence lead time until work starts on a task.

### 3.2.2. New tasks added

The *New Tasks Added* node represents the number of tasks that could be added to the backlog ahead of the task of interest, extending the lead time to backlog exit. As stated earlier, this is a conditional probability dependent on where the team decides to place the task in the backlog. Tasks added to the backlog at a lower priority than the task we are estimating will not influence the lead time, so we need to be mindful of this when analyzing the historical data.

### 3.2.3. Tasks canceled

A Kanban team is constantly re-evaluating whether the items in the backlog still provide value. If those items are no longer relevant, they can be removed from the backlog at any time. The *Tasks Canceled* node expresses how many backlogged tasks will be removed from the backlog ahead of the task of interest. Similar to the addition of tasks, if tasks with a lower priority than the task we are estimating are removed, they will not influence the lead time.

### 3.2.4. Reprioritized backlog

Kanban has the goal of allowing teams to be more flexible or agile in how they perform their work, and one way it achieves this is by allowing the team to change the prioritization of the tasks in the backlog as they learn new information. This could happen through a variety of channels including customer feedback based on previously delivered work content, user research, or sales data. The next uncertainty node, *Reprioritized Backlog*, is a probabilistic representation of how much lower or how much higher the task may be reprioritized, leading to an increase or decrease in the lead time, respectively.

### 3.2.5. Average task size

The *Average Task Size* node represents the work content of the task measured by average story points of the tasks prioritized above the task of interest. Each task in the backlog will require different amounts of work to be completed. This will affect how quickly they move through the workflow, and thus the lead-time for the trailing tasks. Teams often estimate this work content by assigning points to each task, sometimes referred to as story points, with more points signaling higher work content. Since this practice is already common, and teams will likely have this data available, we have incorporated the same technique for quantifying work content into our model.

### 3.2.6. Virtual backlog position

The three uncertainty nodes (*New Tasks Added to Backlog*, *Backlog Tasks Canceled*, and *Task Reprioritized Lower*) and the *Backlog Position* decision node all feed into the *Virtual Backlog Position*. Since the state of this node is determined exclusively by these four nodes, it is a deterministic node represented by a rounded rectangle. The *Virtual Backlog Position* is calculated based on the task's original backlog position, tasks added or removed from the backlog, and changes based on reprioritization. This virtual backlog position is a departure from current story point estimation techniques developed for Scrum, which fail to account for changes to the backlog. Therefore, it is a valid assumption for Scrum to have minimal movement within the sprint backlog. However, Kanban has a continuously dynamic backlog by design, so the same assumption cannot be applied. By modeling with a Bayesian network, we expand the capabilities of story point estimation to work for Kanban in addition to Scrum.

### 3.2.7. Average team daily velocity

A team's velocity is the number of story points they complete on any given day. Since the story points represent the work content of the tasks, velocity represents the pace of work completion. The *Average Team Daily Velocity* node takes into account the rate at which a team is completing work. There are a number of reasons why a team will have variability in their velocity. Since story points are only an estimation, the actual amount of work content may be different than the team expects. Outside

factors could influence the team's productivity, such as members being out sick or on vacation or the burden of work content not accounted for on the Kanban board. The value of this node represents the average velocity of the team over the lead time of the task. Over a longer time period, the probability distribution for average velocity will narrow. We account for this by having the *Virtual Backlog Position* node influence the *Average Team Daily Velocity* node.

### 3.2.8. Days until work starts on task

The final node in the graph is the outcome of the model, the lead time for the task to move from its current location in the backlog to entering the workflow pipeline, and becoming WIP. The *Days Until Work Starts on Task* node is a deterministic node and depends on the *Backlog Project Sizes in Points*, *Virtual Backlog Position*, and *Average Team Velocity*. Although *Lead Time to Exit Backlog* is deterministic, since two nodes pointing to the outcome node are uncertain nodes, the lead time outcome will be probabilistic. The Bayesian network enables the calculation of probabilities for different lead times, instead of the common practice of providing the best case and worst-case estimate.

### 3.2.9. Model assumptions and limitations

Three main assumptions are built into this model. First, the project team is reprioritizing projects on a regular basis. If there are recurring events, such as a weekly backlog grooming meeting where a substantial reprioritization occurs, the accuracy of the model may be affected. Second, it is assumed the team continues to perform in a similar manner as it has historically, due to the model being based on historical data. There may be modifications to the team composition, like a team member being added or removed, or alterations to the subsequent workflow the team uses to complete tasks. An example could be the addition of a rigorous quality control step to the workflow pipeline. Changes like these will affect the underlying data used to generate the associated probability distributions, and new data may need to be collected by the team to re-populate the model. Third, it is assumed the team's work environment does not change. If there are external events that change the characteristics of the tasks in the backlog that were not captured in the historical data, they would not be reflected in the lead time calculations. For instance, if a new product is launched that is filled with critical bugs, the team may get an influx of high priority or large tasks into their backlog. If a similar event is not in the data collected to build the model, the estimates may be inaccurate until the team's environment returns to normal. Because this system is based on historical data, a greater volume of data will improve estimates generated. However, there is a risk that events that will occur are not represented by historical data, and in that case, an outcome may occur to which the system assigned a zero probability.

## 4. Model data and results

### 4.1. Team data collection

In order for the model to provide value to the Kanban team, effective data collection is needed. Data similar to that gathered by teams practicing story point estimation will be able to adequately populate the *Average Backlog Item Size* and *Average Team Daily Velocity* nodes, but more detailed data collection is required for the other nodes. However, an influence diagram allows the user to populate the probabilities in each node using different techniques, including model-based methods or expert-judgment. For example, a distribution could be fit to historical data and used to smooth the distributions entered into the model. Alternatively, the deep learning model suggested by Choetkiertikul et al. (2018) for estimating the number of points in user stories based on natural language processing could be used to populate the story size data actively. If a team does not have data such as how frequently a new task will be added, the team may be able to populate that influence diagram using an expert judgment technique such as the Delphi based planning

poker method frequently used for story point estimation or continuously add new data to the model.

For each uncertainty node in Fig. 3, data was collected from a Kanban team at Andersen Corporation. Data were collected for: the amount of story points assigned to individual tasks, the quantity and location in the backlog of tasks canceled, the quantity and location of reprioritized tasks, the sum of story points completed each day, and the number of new tasks added to the backlog and where they were prioritized. This dataset was collected by recording the backlog each day after a meeting. The dataset was used to calculate conditional probabilities. This manual process was time-consuming and tedious, with each additional node in the model greatly expanding the amount of data that needed to be collected and analyzed for conditional probabilities. This is one reason we suggest a relatively simple model with a minimal number of nodes. There may be value in automating the data collection and calculation of conditional probabilities, reducing the need for teams to manually manipulate data. Another future opportunity could be to model more complex or unknown interactions through the implementation of a learning model. The daily positions of the tasks could be represented as a graph, then processed using a graph neural network to generate probabilistic lead-time estimates in a similar fashion to the system implemented by Park and Park (2019). This method would enable the possibility of reducing data analysis complexity and continuous integration of data for teams using digital Kanban boards such as those available through Asana or Atlassian's Jira platform. The more data available to populate the model, the more accurate the estimates will be. Data were collected for six weeks at Andersen Corporation to estimate the probabilities for each uncertainty seen in Fig. 4.

#### 4.2. Populating the model

The software Netica was used to transform the influence diagram from Fig. 3 into a quantitative model. As seen in Fig. 4, the model

contains the same nodes and relationships, as in Fig. 3. However, Fig. 4 depicts the possible states of each node for a task at position 5 in the backlog. By collecting and monitoring team data and entering the data into each respective node, the distributions are unique to each Kanban team.

For the backlog position decision node, bins with intervals of five were chosen because they provided a balance between providing the team with enough resolution to discuss the estimates with stakeholders while reducing complexity in the data analysis required for the downstream nodes. Because this is a decision node, no probabilities are used, and the user selects the backlog position (5, 10, 15, 20, or 25), with Fig. 4 showing position five selected.

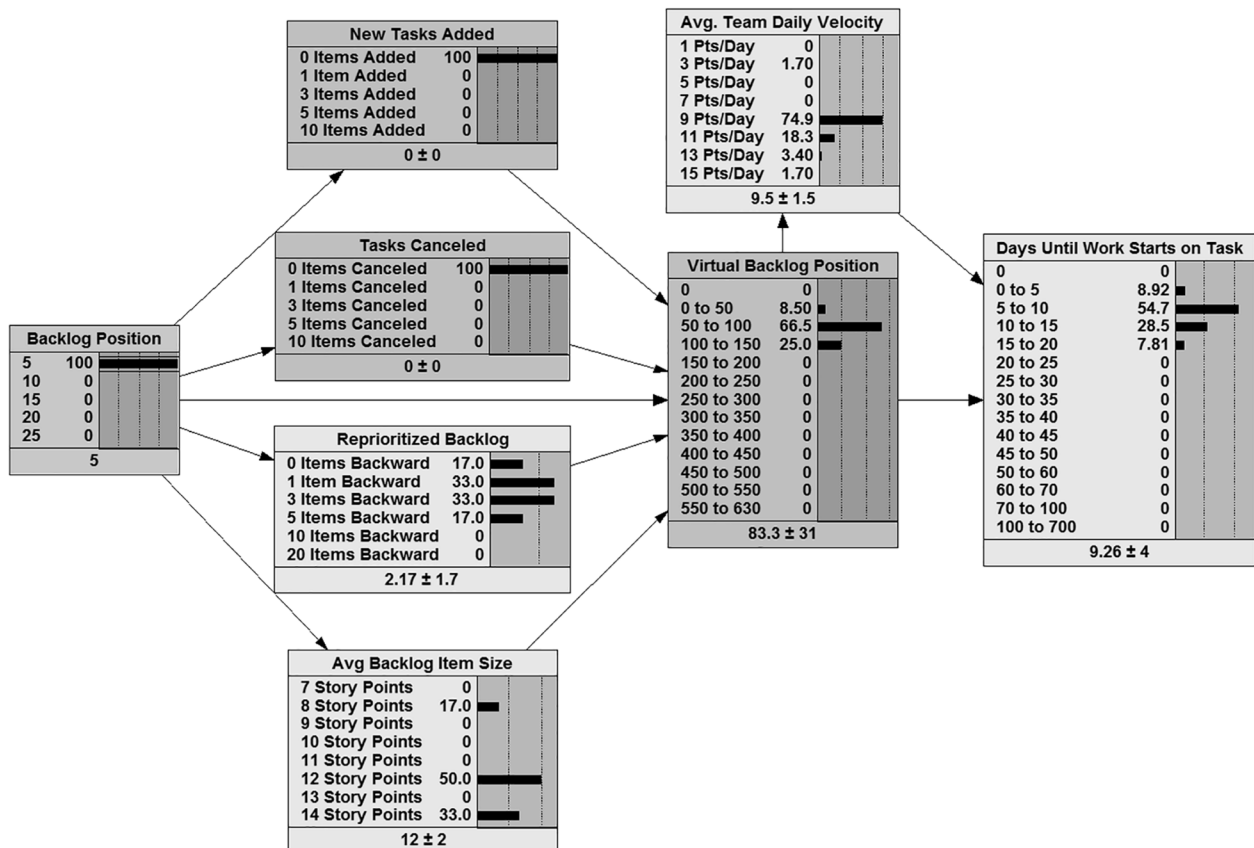
The data for the number of story points given the task's *Backlog Position* was collected for the different backlog positions. Table 1 contains the average number of story points for each project ahead of the new task. The values in this table are in the form of percentages, so there is a 17% probability that a task in the fifth position has four projects ahead of it, each with an average value of story points of eight.

In the duration of our data collection efforts, all of the new tasks added to the backlog were prioritized below task 25, and no tasks were removed from the backlog. This led to the *New Tasks Added* and *Tasks*

**Table 1**

Conditional probabilities for the Average Backlog Item Size. Average of all tasks in the backlog with higher priority than the target task for each backlog position value.

| Backlog Position | 8 Story Points | 10 Story Points | 11 Story Points | 12 Story Points | 14 Story Points |
|------------------|----------------|-----------------|-----------------|-----------------|-----------------|
| 5                | 17             | 0               | 0               | 50              | 33              |
| 10               | 0              | 67              | 33              | 0               | 0               |
| 15               | 0              | 67              | 33              | 0               | 0               |
| 20               | 0              | 33              | 67              | 0               | 0               |
| 25               | 0              | 50              | 50              | 0               | 0               |



**Fig. 4.** Netica Bayesian network for estimating when a backlog task will become WIP displaying network conditions for a task in position five (Weflen, Korniejczuk, Lau, Kryk, MacKenzie, & Rivero, 2018).

*Canceled* nodes both showing a 100% probability that these events will occur zero times. The project team we worked with expressed that this was unusual and that on occasion they will prioritize a task higher in the backlog, and they do sporadically remove tasks. However, these events did not occur in our data collection time period. In this situation, the team could use expert judgment to estimate probabilities for different values, collect more data until these rare events are captured, or they could leave the node in its current condition. We chose the latter in this instance because if these events are rare, they likely do not have much bearing on the estimated lead time for this team. Each project team will have different practices, and there could be teams that would utilize these nodes more heavily.

The states for *Reprioritized Backlog* consist of the number of positions that a task is moved backward, which can be 0, 1, 3, 5, 10, or 20 positions. Because this node has an arrow from the backlog position node, these values consist of probabilities conditioned on the backlog position. Table 2 shows these conditional probabilities. It is apparent that there is a high probability of the task being reprioritized for this team. It is observed that a task prioritized in the fifth position has an 83% probability that other tasks will be reprioritized in front of it before it exits the backlog, while the current story point estimation technique assumes that zero tasks will be moved ahead of the tasks being estimated. In addition, there is a 17% chance that 5–10 items are reprioritized ahead of this task, which means that the current lead time estimates would account for less than half of the work content being completed, resulting in overly optimistic estimates. This is not a large issue for teams using current techniques with Scrum since they hold their backlog mostly constant for the duration of their sprint. However, when the current technique is applied to Kanban teams who reprioritize more frequently, it could increase the inaccuracy of the estimates. The application of a Bayesian network allows us to account for this reprioritization into our initial estimate.

The values for the *Average Task Size* node are presented in Table 3. The values for the average task size ahead of the task we are estimating the lead time for range from 8 to 14 for tasks in position five, but have a narrower range of 10–11 for all other positions. This matches our expectation that as tasks move further down the backlog, the variability will decrease. This data set has distributions that appear sparse and may have benefited from additional data collection. Another option would be to fit a distribution to the data collected or to use expert judgment to fill out the distributions.

The *Virtual Backlog Position* node is calculated using Equation 2. This node estimates the number of story points that will need to be completed before work starts on the new task. This deterministic node simplifies the calculation of the lead time in the final outcome node.

$$\text{VirtualPosition} = (\text{BacklogPosition} + \text{NewTasksAdded} - \text{TasksCanceled}) * \text{AvgTaskSize} \quad (2)$$

The *Average Team Daily Velocity* is the number of story points completed per day on average by the team over the time period that the task is in the backlog and is probabilistically dependent on the *Virtual*

**Table 2**

Conditional probabilities for the Reprioritized Backlog node. Presents reprioritization within the backlog of the target task for each backlog position.

| Backlog Position | 0 Tasks Backward | 1 Task Backward | 3 Tasks Backward | 5 Tasks Backward | 10 Tasks Backward | 20 Items Backward |
|------------------|------------------|-----------------|------------------|------------------|-------------------|-------------------|
| 5                | 17               | 33              | 33               | 17               | 0                 | 0                 |
| 10               | 17               | 0               | 49               | 17               | 17                | 0                 |
| 15               | 17               | 0               | 33               | 17               | 33                | 0                 |
| 20               | 17               | 0               | 17               | 32               | 17                | 17                |
| 25               | 17               | 0               | 17               | 32               | 17                | 17                |

**Table 3**

Conditional probabilities of the Average Task Size node that represent the average number of story points per task

| Backlog Position | 8 Story Points | 10 Story Points | 11 Story Points | 12 Story Points | 14 Story Points |
|------------------|----------------|-----------------|-----------------|-----------------|-----------------|
| 5                | 17             | 0               | 0               | 50              | 33              |
| 10               | 0              | 67              | 33              | 0               | 0               |
| 15               | 0              | 67              | 33              | 0               | 0               |
| 20               | 0              | 33              | 67              | 0               | 0               |
| 25               | 0              | 50              | 50              | 0               | 0               |

**Table 4**

Conditional probabilities for the Virtual Backlog Position node. Represents the team's average velocity per day for each range of virtual backlog positions.

| Virtual Backlog Position | 3 Points per Day | 9 Points per Day | 11 Points per Day | 13 Points per Day | 14 Points per Day |
|--------------------------|------------------|------------------|-------------------|-------------------|-------------------|
| 0                        | 0                | 100              | 0                 | 0                 | 0                 |
| 1–50                     | 20               | 0                | 20                | 40                | 20                |
| 51–100                   | 0                | 75               | 25                | 0                 | 0                 |
| 100–630                  | 0                | 100              | 0                 | 0                 | 0                 |

*Backlog Position*. As seen in Table 4, these probabilities are conditional upon the virtual backlog position since the distribution of the average will narrow if the task is further back in the backlog.

To determine how many days until work begins on a backlogged item, the team sets the value of the decision node *Backlog Position*. By entering the decision made by the team of where to prioritize the task in the backlog into this node, the distribution of probabilities for the lead time is calculated using Equation 3 in the final outcome node *Days Until Work Starts on Task*. Even though the lead time for a task to exit the backlog is a deterministic function of two variables, *Average Team Daily Velocity* is an uncertainty and *Virtual Position* is derived from other uncertainties, meaning the lead time will also be uncertain.

$$\text{Lead Time} = \frac{\text{Virtual Position}}{\text{Average Daily Velocity}} \quad (3)$$

Fig. 4 shows the model for a task prioritized at the fifth position in the backlog. If the user were to select the tenth position in the decision node, the distributions in each of the downstream nodes, including the lead times, would change for this position. This output can be utilized by the project team to communicate to customers an estimate of when work will start on their requested task, or decide where to place a task within the backlog. The Netica model in Fig. 4 shows the probability of each lead time as a probability mass function or histogram, which may not be

the most helpful way for a user to think about lead times. We will next present these results in the form of a cumulative distribution function (CDF) and discuss tactics for communicating and understanding this representation.

#### 4.2.1. Lead time estimates

Utilizing the probabilities provided at the final consequence node by fixing the backlog position each of the decision points: 5, 10, 15, 20, and 25, five discrete CDFs were created and are presented in Fig. 5. The x-axis indicates the number of days until the task starts. The y-axis indicates the probability that the task begins within the number of business days specified by the x-axis.

Fig. 5 shows that as the position in the backlog increases, the lead time for the task to exit the backlog increases. Since more tasks are required to be completed before work on the task of interest begins, the lead time is expected to increase as the backlog position increases. The results also demonstrate the uncertainty that is inherent when placing a task in the backlog. For example, if a customer requires work to start on a task within 26 days, placing the task in position five would ensure 100% probability that the task will be on time while placing the task in position 15 would ensure an 80% probability of being delivered on time, and position 25 would only provide a 10% probability that work will start on the task in the desired timeframe.

The stakeholder's risk tolerance around the completion date for the task must be balanced against competing demands on the team and the relative importance of other tasks in the backlog. The team and stakeholder can use the information presented in Fig. 5 to guide their discussion around the decision of where in the backlog the task should be placed. The project team may be able to produce risk tolerance thresholds for certain customers or for classifications of tasks. For example, this team may set a guideline that quality issues should be completed within 21 days, and set a threshold of 90% probability of completion in this timeframe. Fig. 5 can be referenced to see that this team would need to place these quality issues no further back than position 10 in the backlog in order to meet these criteria. Different lead times and thresholds for completion could be set for different types of tasks. If there is a requirement that safety issues are completed within 12 days, safety issues would need to be placed no further back than position 5 in order to ensure at least a 90% chance of meeting this requirement. Putting documentation issues in position 20 may be appropriate because there is a 70% probability that these issues will be completed within 31 days. The actual values for these thresholds would depend on the environment of the team and can be managed on a case by case basis. By doing this, the team could automate some decisions around the placement of certain types of tasks, in turn reducing the effort and time required for making decisions.

This influence diagram may also be used to provide updated estimates to stakeholders as their tasks progress through the backlog. A team may set an automated update to task stakeholders when milestones

in the backlog are reached. If a task starts in position 25, the team may communicate that there is an 80% probability that work will start on that task within 40 days. When that task reaches position 15 in the backlog, an update could be sent informing the interested parties that the task now has an 80% probability that work will start on that task within 25 days. These dynamic updates may allow more transparency to progression through the backlog and allow for more informed discussions between stakeholders and the project team with respect to backlog reprioritization.

#### 4.3. Sensitivity analysis

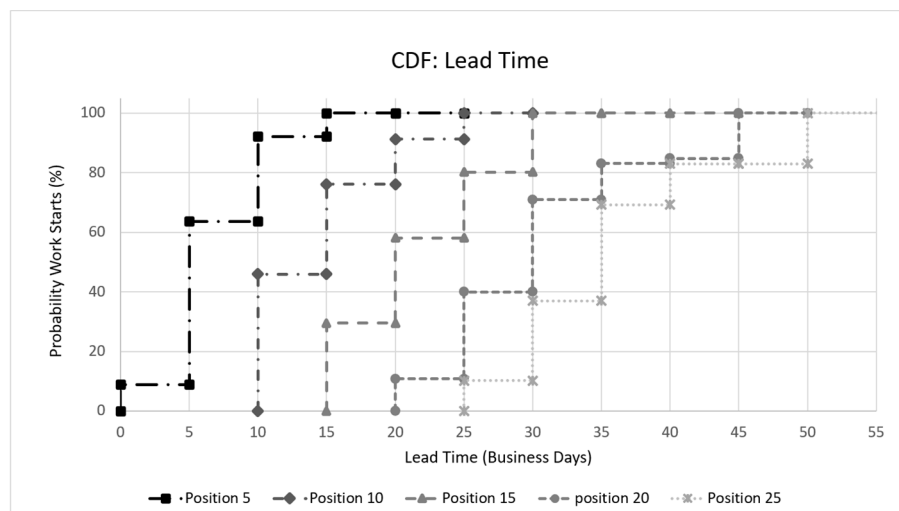
It can be helpful for a team to understand how sensitive the lead time is to changes in the values of the nodes in the influence diagram. The sensitivity analysis will provide the team with a better sense of the accuracy of their estimates as variables change. For example, if a team member is sick for a week, it would be useful to understand if this will lead to a major or minor change in the actual lead times for tasks. In addition, knowing which factors have a large influence can help us determine where it may be helpful to collect more data in order to improve our forecasts.

A sensitivity analysis is conducted to determine how the 80% quantile changes as variables in the influence diagram change. In the base-case situation, when the backlog position node is set at position 15, the 80% quantile equals 30 days. This means that there is an 80% probability the task will begin in 30 days or less. Each of the five uncertainty nodes is varied one at a time to a minimum and maximum values, and the 80% quantile for the lead time is calculated for each of these minimum and maximum values, as seen in Table 5. Fig. 6 depicts these results in a tornado diagram when the task is held constant at position 15.

Fig. 6 shows that our Bayesian network model of Agile Kanban backlog lead-times is most sensitive to the team's average velocity, especially if the average velocity is at its lowest value of 3 points per day.

**Table 5**  
Minimum and maximum values for each uncertainty node.

| Node                     | Minimum Value | Maximum Value |
|--------------------------|---------------|---------------|
| New Tasks Added          | 0             | 10            |
| Tasks Canceled           | 0             | 10            |
| Reprioritized Backlog    | 0             | 10            |
| Average Task Size        | 7             | 14            |
| Avg. Team Daily Velocity | 3             | 15            |



**Fig. 5.** Discrete cumulative distribution function. lead time as a function of the backlog position.

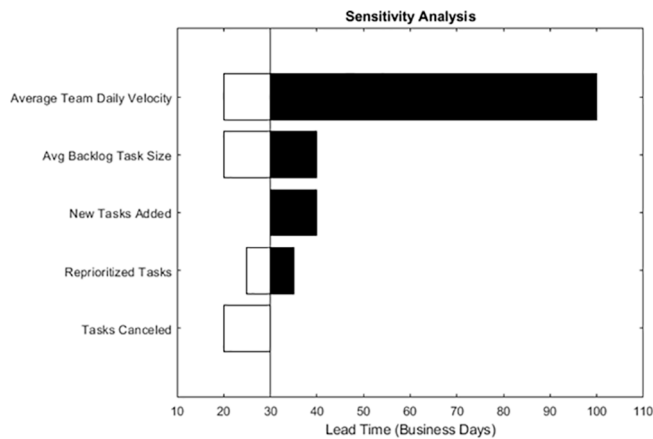


Fig. 6. Lead time for work to start on a task in position 15 of the backlog with an 80% probability.

If the average daily velocity is 3 points per day, the 80% quantile for the lead time is more than three times as large as the nominal value of 30 days. The average velocity could be this low if team members are pulled onto different projects, are out sick, or if the size of the team is adjusted. This information can be used by a team to better set stakeholder expectations for delivery if these or other events are anticipated to occur. Fig. 6 shows the 80% quantile for lead time changes more when the team's velocity is moved to its highest level than when the team's velocity is moved to its lowest level. The model is most sensitive to the value of this node, suggesting that it would be valuable to have accurate projections of a team's velocity when creating estimates.

The average size of each task, measured in story points, is varied between 7 and 14. This results in the second largest range of lead times. If a team finds that the estimates are not matching the actual lead time values, it will make sense for the team to gather better data for the *Average Task Size* node after the *Average Team Daily Velocity* node.

If a team is more interested in using this Bayesian network to improve the accuracy of their estimates, instead of reducing the effort, an option would be to continue to estimate the work content of all backlogged tasks using planning poker or another method of their choice. They could enter their planning poker estimated value into the *Average Task Size* node, and the remainder of the model would update accordingly. This could be conducted on an ongoing basis for all estimates, or it could be conducted on a single occasion when additional accuracy is required for a lead-time estimate. Other methods for generating estimates for the work content could use natural language processing and deep learning to gain insights into the work content of tasks in the Backlog (Choetkiertikul et al., 2018). The values given by this model could be entered into the *Average Task Size* node. The model we have created is flexible because it can take outputs from other methods that generate values for each node. These methods could include historical data, expert elicitation, simulation, and machine learning.

#### 4.4. Comparison with current prediction techniques

While the several techniques for producing lead time estimates were discussed in the introduction section, this study will directly compare the new method to the ones commonly used by Kanban teams. The primary method used for measuring lead time in Kanban involves tracking the number, or flow, of tasks completed during a time interval, such as daily or weekly, and calculate an average task completion rate (TCR), which can be used to approximate the time that a task in the backlog will be completed (Al-Baik & Miller, 2015; Anderson, 2011). This method can be taken a step further by also measuring the tasks added to the backlog and calculating an average task addition rate

(TAR), and subtracting this from the TCR (Brechtner, 2015). However, this method assumes that all tasks being added to the backlog are prioritized ahead of the task of interest, which may be far too pessimistic for some teams. During the observation period of the target project team, they achieved an average TCR of 2.0 tasks per day and an average TAR of 1.7 tasks per day. These values are likely similar because the observed team is fulfilling a sustained engineering role. A Kanban team completing a finite list of features for a product release would expect a bigger gap between the TCR and the TAR, with the TAR being the lower value.

As Table 6 demonstrates, the TCR provides an optimistic prediction while subtracting out the TAR results in a pessimistic prediction when compared to the CDF produced by the influence diagram (Fig. 5). A key differentiator between the estimates produced using current techniques and the new method presented in this study is that current methods lack a depiction of the uncertainty in the estimates. Agile project management is often used in environments with high uncertainty with many factors, both technical and non-technical, that can affect lead time estimates (Badampudi, Fricker, & Moreno, 2013). Management of uncertainty is critical to project success, and for less rigidly defined projects with 'softer' scopes, the team must manage stakeholder expectations and tolerance of uncertainty (Atkinson, Crawford, & Ward, 2006). The expert system presented in this study will generate a cumulative distribution function that the project team can use to communicate uncertainty to project stakeholders. In addition, this system gives the team the flexibility to test multiple scenarios by adjusting the values in the nodes of the influence diagram.

While uncommon in practice, several methods have been used to approximate the uncertainty in the lead time predictions calculated using TCR and TAR by producing three-point estimates that evaluate a pessimistic, average, and optimistic lead time prediction (Coelho & Basu, 2012; Zia, Kamal, & Ziauddin, 2012). While the three-point estimates on their own can demonstrate an understanding that there is a range of possible outcomes, they do not represent the likelihood of a certain outcome. However, a probability distribution, such as a triangular or beta distribution, can be paired with these estimates to model the uncertainty of the events using tools like Program Evaluation and Review Technique (PERT) or Monte Carlo simulations (Hannay, Benestad, & Strand, 2019). To demonstrate this technique, the velocity for the observed Kanban team was averaged for each week to produced best case, worst case, and average lead time estimates (Table 6). It can be seen that the predicted average lead time calculated using story points differs from those produced when simply looking at the task completion rate in Table 5. This shows how the variability in effort for each task can affect the lead time (See Table 7).

The three-point estimates were used to generate the triangular probability distributions in Fig. 7a. These functions can be used to generate a CDF (Fig. 7b) for the lead time estimates, similar to that produced by the Influence Diagram in Fig. 5. While the CDFs produced using this method appear similar to those from the influence diagram, there are several reasons why they are limited. First, the only uncertainty that was accounted for was that of the team productivity, measured by the velocity metric. While this is a key driver of variability in Scrum teams, Kanban teams are also affected by the addition, cancelation, and reprioritization of tasks. The sensitivity analysis

Table 6

Lead time estimates based on average TCR and TAR for the observed Kanban team.

| Backlog Task Position | TCR Lead Time Estimate (Days) | TCR – TAR Lead Time Estimate (Days) |
|-----------------------|-------------------------------|-------------------------------------|
| Position 5            | 2.5                           | 15                                  |
| Position 10           | 5                             | 30                                  |
| Position 15           | 7.5                           | 45                                  |
| Position 20           | 10                            | 60                                  |
| Position 25           | 12.5                          | 75                                  |

**Table 7**

Lead time estimates calculated using best case, average and worst case story point velocity for the observed Kanban team.

| Backlog Task Position | Best Case Lead Time (Days) | Average Lead Time (Days) | Worst Case Lead time (Days) | PERT Expected Lead Time (Days) |
|-----------------------|----------------------------|--------------------------|-----------------------------|--------------------------------|
| Position 5            | 2.5                        | 3.3                      | 11.5                        | 5.8                            |
| Position 10           | 3.3                        | 4.4                      | 15.1                        | 7.6                            |
| Position 15           | 6.5                        | 8.8                      | 30.2                        | 15.2                           |
| Position 20           | 7.5                        | 10.1                     | 34.5                        | 17.4                           |
| Position 25           | 10.4                       | 14.0                     | 48.0                        | 24.1                           |

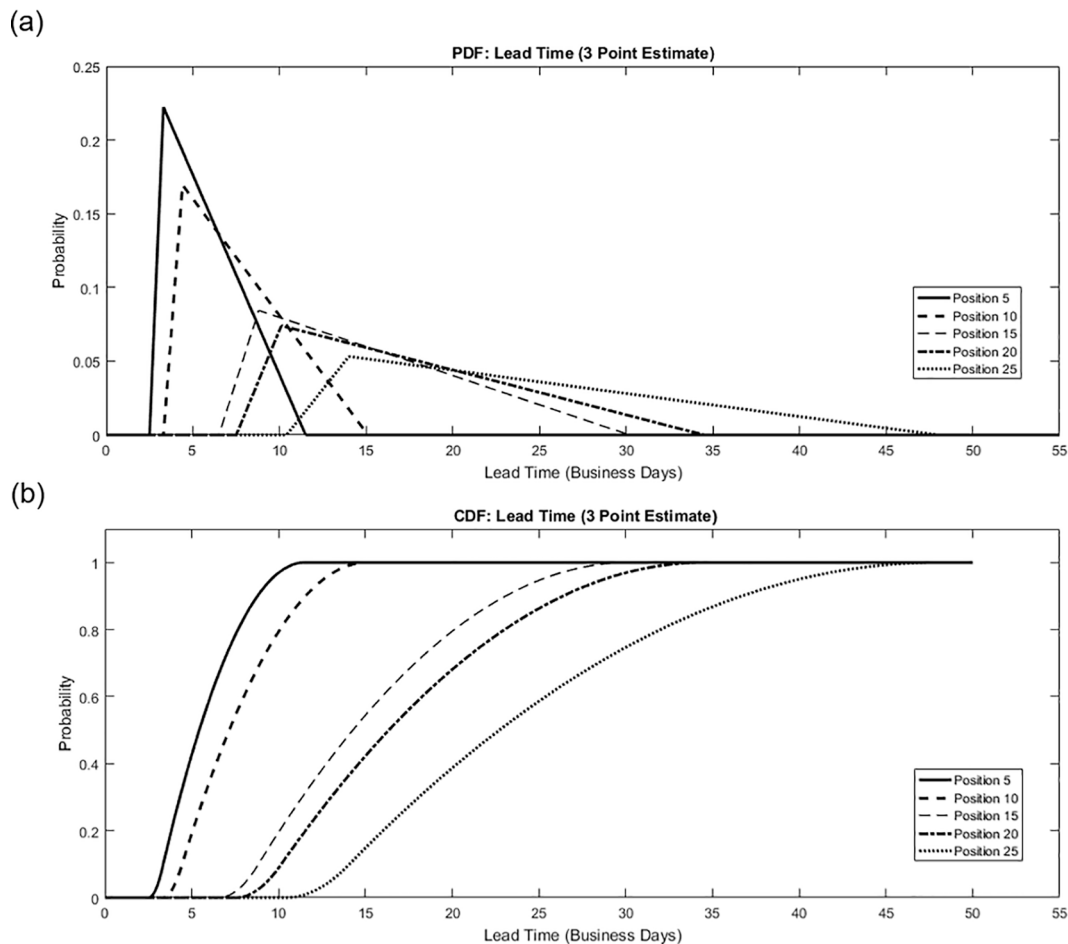
revealed that together these influencing factors have a bias toward increasing the lead time. This would mean that estimates produced using current methods would be optimistic and more narrowly distributed than those produced by the influence diagram. This is apparent when comparing the CDF for a task in position 25 for the two models, with the Bayesian network estimating zero probability that the task will be completed in less than 25 days and the current model giving that same lead time a high probability. Optimistic estimates may result in unsatisfied stakeholders when tasks are delivered late compared to the estimated delivery. Additionally, these estimates require the project team to continue estimating task effort using planning poker to assign story points. Time spent on effort estimation by the team could be directed toward making progress on delivering value by working on tasks. While this can be overcome by using a simpler metric like TCR and TAR, the accuracy of the estimate may suffer. The triangular distribution, or any other assumed distribution, may not accurately capture the true

uncertainty and variability in the factors. Calculating probabilities based on data that the team has collected and using those probabilities in the influence diagram can potentially match the uncertainty and variability in the influencing factor more closely than assuming a distribution. The influence diagram provides a graphical explanation of the factors that are influencing the lead time, which may help the team explain the causes of uncertainty to stakeholders and to conduct what-if analysis with ease. The explainable nature of the expert system using an influence diagram may help increase stakeholder confidence in predictions.

The objective of this comparison between the proposed expert system and current methods is not to demonstrate that the new method provides superior estimates. To accomplish this, a more thorough analysis would be needed with a larger data set and data from different project teams. Instead, this comparison shows that the proposed method both demonstrates uncertainty in the lead time estimates and in the factors that influence the estimate. These uncertainties are accounted for along with their interdependencies in an explainable format that can help teams better communicate with stakeholders while managing their expectations.

## 5. Conclusions

The objective of this paper was to provide an alternative approach to lead time estimation for Agile Kanban project teams to account for the addition and removal of tasks to the backlog, their reprioritization, and the team's level of productivity. This research acts as an example of how influence diagrams can be incorporated into an expert system to make the system more explainable and to communicate uncertainty of the



**Fig. 7.** Probability Density Functions (PDF) and Cumulative Distribution Functions (CDF) for task lead times using story point velocity data from the observed Kanban team.

system outputs. Automating this process, through the creation of an expert system, can reduce the level of effort required by the team to produce these estimates. An influence diagram representing the factors impacting the lead time was created and used to create an interactive tool using the software Netica. The probabilistic output of this tool is represented with a CDF graph that can be used by a project team to visually communicate lead time probabilities to stakeholders or to use as a reference during the decision-making process of backlog prioritization.

A project team can use the probabilistic representation of the estimated lead time to explain to stakeholders that while there is a high probability that a task in this position is completed within a few days, there is a chance the actual lead time is longer. This can help teams better manage stakeholder expectations. The ability to set expectations in this way is critical in the project management profession, which is driven by the criticality of meeting customers' demands on-time. Sensitivity analysis demonstrated that the estimated lead time is most sensitive to the team velocity and the work content of the tasks.

The estimates produced in this paper are only as accurate as historical data on which they are based. Having a larger data set will capture more of the rare events and should improve the accuracy of the conditional probabilities within the influence diagram. While further investigation with a larger data set is required to evaluate the quality of the estimates produced by this expert system compared to current methods, the presented system's true advancement is a depiction of the uncertainty associated with those estimates in an explainable format. A team implementing this tool must be sure that they have sufficient data to achieve a level of accuracy in the estimates that meet their requirements. If the team experiences change in the team members, the type and size of tasks being assigned, or the way tasks are being prioritized, then the model will need new data collection and the probability distributions updated. The tool could be further improved through the implementation of automating data collection and integration. Teams that are using a digital system to track their tasks could implement this automation. The conditional probabilities in the nodes in the influence diagram could be populated through other means, such as through the integration of a natural language processing model to estimate the average backlog item size. This may allow for the incorporation of real-time data analytics in conjunction with the historical team data.

This research was motivated by the focus on continuous improvement in both Lean and Agile, and with the goal of contributing to the continued development of Kanban. By developing an expert system that enables the communication of probabilistic lead time estimates, teams can better manage stakeholder expectations. The team members can use the outcome of this model to gain better insight into the impact of their decision on where to prioritize a task in the backlog. Expert systems utilizing influence diagrams that account for complex interactions in order to manage stakeholder expectations do not need to be limited to Kanban or even Agile project management. These concepts can be applied more broadly in the production of estimates and risk assessment within the field of project management. This paper opens opportunities to improve the effectiveness and performance of project managers and project teams by automating work through the creation of other expert systems. With further evaluation and more extensive data sets this method can be evaluated with other Agile project management techniques such as Scrum and Scrumban. By demonstrating the ability to use historical data from a team practicing Kanban to populate and influence diagram based expert system in Netica, we open the possibility for other researchers or practitioners to implement and improve project management in an Agile environment. The ability of influence diagrams to use limited data sets and expert elicitation to communicate uncertainty in an explainable format can be applied to expert systems beyond just those used in project management.

#### CRedit authorship contribution statement

**Eric Weflen:** Conceptualization, Methodology, Writing – original

draft. **Cameron A. MacKenzie:** Formal analysis, Writing - review & editing. **Iris V. Rivero:** Writing - review & editing, Supervision.

#### Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eric Weflen had a prior professional relationship with Andersen Windows.

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